

3. Develop a Django app that performs student registration to a course. It should also display list of students registered for any selected course. Create students and course as models with enrolment as ManyToMany field.

#settings.py

```
INSTALLED_APPS = [  
    'django.contrib.admin',  
    'django.contrib.auth',  
    'django.contrib.contenttypes',  
    'django.contrib.sessions',  
    'django.contrib.messages',  
    'django.contrib.staticfiles',  
    'courses',  
]
```

#urls.py

```
from django.contrib import admin  
from django.urls import path, include  
  
urlpatterns = [  
    path("admin/", admin.site.urls),  
    path('', include('courses.urls')), # Include the app URLs  
]
```

#courses/views.py

```
from django.shortcuts import render, redirect, get_object_or_404  
from .models import Student, Course  
from .forms import StudentRegistrationForm, CourseSelectionForm  
  
def register_student(request):  
    if request.method == 'POST':  
        form = StudentRegistrationForm(request.POST)  
        if form.is_valid():  
            student = form.save()  
            return redirect('select_course', student_id=student.id)  
    else:  
        form = StudentRegistrationForm()  
        return render(request, 'courses/register_student.html', {'form': form})  
  
def select_course(request, student_id):
```

```

student = get_object_or_404(Student, id=student_id)
if request.method == 'POST':
    form = CourseSelectionForm(request.POST)
    if form.is_valid():
        course = form.cleaned_data['course']
        course.students.add(student)
        return redirect('course_students', course_id=course.id)
    else:
        form = CourseSelectionForm()
    return render(request, 'courses/select_course.html', {'form': form,
'student': student})

def course_students(request, course_id):
    course = get_object_or_404(Course, id=course_id)
    students = course.students.all()
    return render(request, 'courses/course_students.html', {'course': course,
'students': students})

```

#Courses/urls.py

```

from django.urls import path
from . import views

urlpatterns = [
    path('register/', views.register_student, name='register_student'),
    path('select_course/<int:student_id>/', views.select_course,
name='select_course'),
    path('course_students/<int:course_id>/', views.course_students,
name='course_students'),
]

```

#courses/models.py

```

from django.db import models

class Student(models.Model):
    first_name = models.CharField(max_length=50)
    last_name = models.CharField(max_length=50)
    email = models.EmailField(unique=True)

```

```

    def __str__(self):
        return f"{self.first_name} {self.last_name}"

class Course(models.Model):
    name = models.CharField(max_length=100)
    description = models.TextField()
    students = models.ManyToManyField(Student, related_name='courses')

    def __str__(self):
        return self.name

```

#courses/forms.py

```

from django import forms
from .models import Student, Course

class StudentRegistrationForm(forms.ModelForm):
    class Meta:
        model = Student
        fields = ['first_name', 'last_name', 'email']

class CourseSelectionForm(forms.Form):
    course = forms.ModelChoiceField(queryset=Course.objects.all(), required=True,
label="Select a Course")

```

#courses/admin.py

```

# courses/admin.py

from django.contrib import admin
from .models import Student, Course

class StudentAdmin(admin.ModelAdmin):
    list_display = ('first_name', 'last_name', 'email')
    search_fields = ('first_name', 'last_name', 'email')

class CourseAdmin(admin.ModelAdmin):
    list_display = ('name', 'description')
    search_fields = ('name',)
    filter_horizontal = ('students',)

admin.site.register(Student, StudentAdmin)

```

```
admin.site.register(Course, CourseAdmin)
```

#templates\courses\course_students.html

```
<!-- courses/templates/courses/course_students.html -->

<!DOCTYPE html>
<html>
<head>
    <title>Students Enrolled in {{ course.name }}</title>
</head>
<body>
    <h1>Students Enrolled in {{ course.name }}</h1>
    <ul>
        {% for student in students %}
            <li>{{ student.first_name }} {{ student.last_name }}</li>
        {% endfor %}
    </ul>
    <a href="{% url 'register_student' %}">Register another student</a>
</body>
</html>
```

#templates/courses/register_student.html

```
<!-- courses/templates/courses/register_student.html -->

<!DOCTYPE html>
<html>
<head>
    <title>Register Student</title>
</head>
<body>
    <h1>Register Student</h1>
    <form method="post">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Register</button>
    </form>
</body>
</html>
```

#courses/templates/courses/select_course.html

```
<!-- courses/templates/courses/select_course.html -->

<!DOCTYPE html>
<html>
<head>
    <title>Select Course</title>
</head>
<body>
    <h1>Select Course for {{ student.first_name }} {{ student.last_name }}</h1>
    <form method="post">
        {% csrf_token %}
        {{ form.as_p }}
        <button type="submit">Enroll</button>
    </form>
</body>
</html>
```